

Quantum phase transitions

The dynamical critical exponent is introduced. The phase diagram and the phase transitions in the Bose–Hubbard model of interacting bosons on a lattice are determined. The concept of quantum fluctuations is introduced on the example of an interacting superfluid, and finally the special scaling of conductivity is discussed.

8.1 Dynamical critical exponent

The finite temperature phase transitions studied in previous chapters are the result of the competition between the entropy and the energy terms in the free energy: the weight of entropy increases with temperature, and ultimately destroys the order that may be existing in the system. A sharp phase transition between two phases exhibiting qualitatively different correlations may, however, occur even at zero temperature, by varying a coupling constant in the Hamiltonian. The transition then corresponds to a non-analyticity of the energy of the ground state. A simple example is provided by the interacting bosons, where the superfluid transition may be brought about by tuning the chemical potential at $T = 0$. Such $T = 0$ phase transitions are called *quantum phase transitions*, and will be the subject of the present chapter.

In general, a quantum phase transition may lie at the end of a line of thermal phase transitions, as in the bosonic example mentioned above. It is possible, however, that the system may not even have an ordered state at finite temperatures, but still exhibits a quantum critical point. Two different situations are depicted in Fig. 8.1. Examples of such phase diagrams are provided by the system of interacting bosons which will be discussed in Sections 8.2 and 8.3.

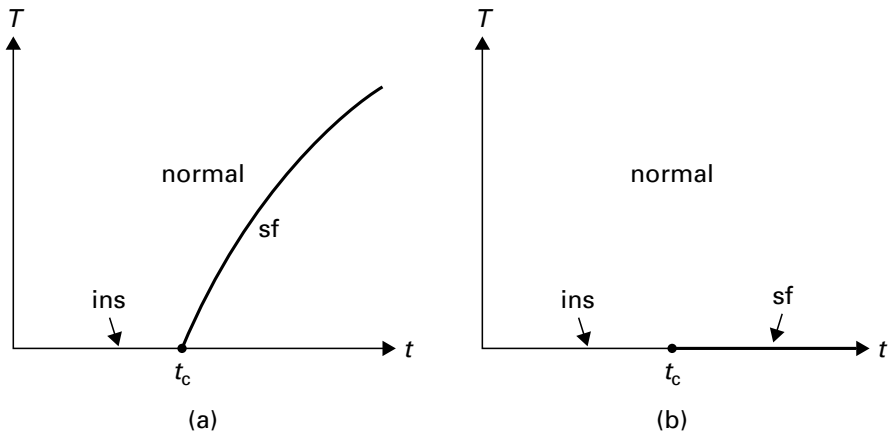


Figure 8.1 Phase diagram of interacting bosons at a density commensurate with the lattice in two (a) and one (b) dimension. The parameter t measures hopping between the nearest-neighboring sites. (a) The system has the Kosterlitz–Thouless transition at $T \neq 0$, and the three-dimensional XY quantum phase transition at $T = 0$. (b) The system exhibits the Kosterlitz–Thouless transition between the insulating and the superfluid phases at $T = 0$ and $t = t_c$, and no phase transition at $T \neq 0$.

As mentioned in the Section 2.3, the principal difference between the quantum ($T = 0$) and the thermal ($T \neq 0$) phase transitions is that while dynamics is irrelevant for the latter, it is crucial for the former. Put differently, whereas near a finite temperature transition the fluctuating order parameter may be considered completely independent of the imaginary time, this is no longer possible in the quantum case. Besides the usual fluctuations in space, the fluctuations in imaginary time need to be taken into account as well, and thus the imaginary time effectively plays the role of an additional dimension of the system. As the system's dimensionality is the crucial factor in determining its critical behavior, it is obviously important to determine how exactly the imaginary time should be counted relative to spatial dimensions. This brings us to the notion of an additional critical exponent at quantum phase transitions, called the *dynamical critical exponent*.

Let us reconsider the superfluid susceptibility in Eq. (3.1). At $T = 0$ it will in principle be a function of the wavevector $\vec{k}$, the chemical potential μ , the interaction λ , and the Matsubara frequency ω , i.e.

$$\chi(\vec{k}, \omega) = F(k, \omega, \mu, \lambda). \quad (8.1)$$

The susceptibility is, of course, still implicitly dependent on the ultraviolet cutoff Λ , and renormalizability implies that

$$\chi(\vec{k}, \omega) = b^x F(bk, b^z \omega, \mu(b), \lambda(b)), \quad (8.2)$$

in analogy with Eq. (3.4). The power x and the functions $\mu(b)$ and $\lambda(b)$ will in principle be different than at $T \neq 0$. Most importantly, the Matsubara frequency and the wavevector may be rescaled differently. The exponent z that quantifies this difference, and which may or may not be unity, is the dynamical critical exponent.

Assume that for $b \gg 1$, $\mu(b) \approx \mu b^{1/\nu}$ and $\lambda(b) \approx \lambda^*$, where ν and λ^* characterize the quantum critical point in the system. Choosing again the parameter b so that $\mu b^{1/\nu} = \mu_0$ with μ_0 constant, the susceptibility at $T = 0$ may be written as

$$\chi(\vec{k}, \omega) = \left| \frac{\mu_0}{\mu} \right|^{x\nu} F(k\xi, \omega\tau, \mu_0, \lambda^*), \quad (8.3)$$

where $\xi = |\mu_0/\mu|^\nu$ is the correlation length, and $\tau = \xi^z$ the *correlation time*. The dynamical exponent therefore may be understood as connecting the characteristic time scale and length scale for fluctuations near the quantum critical point.

The reader may be wondering about the relevance of the exponent z in experiments, which, of course, are always performed at finite temperatures. Consider therefore the finite temperature susceptibility, satisfying a similar scaling equation to Eq. (8.2):

$$\chi(\vec{k}, \omega) = b^x F(bk, b^z \omega, T(b), \mu(b), \lambda(b)). \quad (8.4)$$

As the inverse of temperature enters only as the upper limit of integration in imaginary time, temperature must be rescaled exactly inversely to imaginary time. This therefore is the same scaling as the frequency. So, $T(b) = b^z T$, and the dynamical critical exponent also governs the finite temperature scaling near the quantum critical point. In particular, the uniform static susceptibility is then

$$\chi(0, 0) = \left| \frac{\mu_0}{\mu} \right|^{x\nu} F\left(0, 0, T \left| \frac{\mu_0}{\mu} \right|^{z\nu}, \mu_0, \lambda^*\right), \quad (8.5)$$

and the critical temperature is therefore

$$T_c = c \left(\frac{\mu}{\mu_0} \right)^{z\nu}, \quad (8.6)$$

with c as a non-universal constant with units of temperature. The dynamical critical exponent therefore directly determines the form of the phase boundary in the $\mu - T$ plane near the quantum critical point at $T = 0$. Right at the quantum critical point at $\mu = 0$, on the other hand, we may choose $b^z T = T_0$, where T_0 is some fixed temperature. Then

$$\chi(\vec{k}, \omega) = \left(\frac{T_0}{T}\right)^{x/z} F\left(k\left(\frac{T_0}{T}\right)^{\frac{1}{z}}, \omega\left(\frac{T_0}{T}\right), T_0, 0, \lambda^*\right), \quad (8.7)$$

and the temperature itself provides the characteristic length and time scales for fluctuations.

Since the effective dimensionality of the system at $T = 0$ is always increased by the addition of the imaginary time direction, it often happens that in physical dimensions quantum critical behavior becomes trivial, although the thermal critical behavior at the same transition was not. The reader may thus legitimately worry whether, this being the case, quantum phase transitions require a separate chapter in a book on critical phenomena. We believe the answer is yes, nevertheless, for at least two reasons. First, when the quantum critical behavior is non-trivial, quantum phase transitions often are qualitatively different from the thermal phase transitions, due to the special role played by the imaginary time. Examples of this will be provided in the subsequent sections. Second, in the experiments one is of course interested in the dynamics of the system in real and not imaginary time. This in principle is computable from the response known in the imaginary time, or at the Matsubara frequencies, by analytically continuing to real time, or equivalently to real frequencies. The step of analytical continuation, however, may be highly non-trivial when one knows the function to be continued only approximately. Some, maybe deceptively simple, examples of this will also be discussed in the subsequent sections.

8.2 Quantum critical point in Φ^4 -theory

As a paradigmatic example of a quantum phase transition let us consider a general Φ^4 action given by

$$S = \int_0^{1/T} d\tau \int d\vec{r} [u \Phi^*(\vec{r}, \tau) \partial_\tau \Phi(\vec{r}, \tau) + v |\partial_\tau \Phi(\vec{r}, \tau)|^2 + |\nabla \Phi(\vec{r}, \tau)|^2 - \mu |\Phi(\vec{r}, \tau)|^2 + \lambda |\Phi(\vec{r}, \tau)|^4]. \quad (8.8)$$

To be specific we assumed that $\Phi(\vec{r}, \tau)$ is complex, but an analogous action may be written for a field with an arbitrary number of real components, and

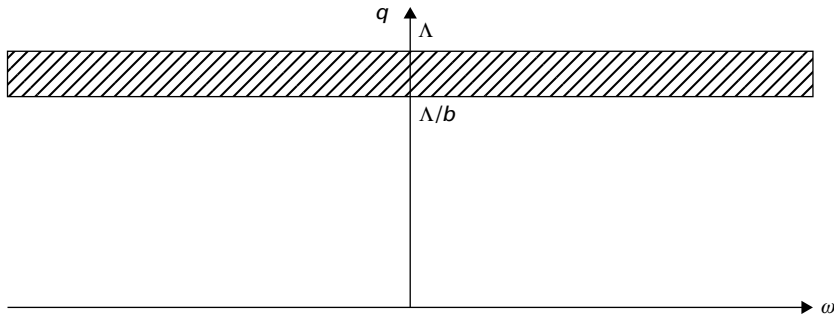


Figure 8.2 The shaded region represents the high-energy modes that are integrated out in the renormalization procedure at $T = 0$.

in particular for the Ising and the Heisenberg universality classes. The τ -dependence of Φ may be neglected at the finite temperature transition, and the universality class is then as described in Chapter 2. At $T = 0$, however, the first two terms must be taken into account, and as a result the universality class of the quantum phase transition will be different from that of its thermal counterpart.

When $u \neq 0$, $v = 0$, and Φ is complex, the above action describes the Bose–Einstein condensation at $T = 0$. For $u = 0$, $v \neq 0$, and a complex Φ , it describes the superfluid–Mott insulator transition in a system of lattice bosons at a commensurate density, as will be discussed in detail in the next section. When $u = 0$ and Φ has one or three real components, S describes the quantum phase transition between magnetically ordered and disordered states of the quantum versions of the Ising and the Heisenberg model, respectively.

Let us assume first that $u = 0$ and that Φ is complex, and determine the flow of the coupling constants at $T = 0$ under the change of cutoff as $\Lambda \rightarrow \Lambda/b$. Following exactly the same procedure as in Section 3.2, integrating the modes with $\Lambda/b < \vec{k} < \Lambda$ and $-\infty < \omega < \infty$, as depicted in Fig. 8.2, to the leading order in λ yields

$$\mu(b) = b^2(\mu - 4\lambda \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{\Lambda/b}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \left(\frac{1}{\omega^2 + k^2 - \mu} - \frac{1}{\omega^2 + k^2} \right), \quad (8.9)$$

$$\lambda(b) = b^{4-d-z}\lambda \left(1 - 10\lambda \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{\Lambda/b}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(\omega^2 + k^2 - \mu)^2} \right), \quad (8.10)$$

where for convenience we have set $v = 1$. The dynamical exponents are determined by the requirement that after rescaling of momenta as $bk \rightarrow k$ and of frequencies as $b^z\omega \rightarrow \omega$ one can rescale the field Φ to bring the action into the

old form. In the present case frequency and momentum appear symmetrically, and therefore

$$z = 1. \quad (8.11)$$

Defining then $\epsilon = 3 - d$, and the dimensionless coupling $\hat{\lambda} = \lambda \Lambda^{d-3} S_d / (4(2\pi)^d)$, the flow equations become identical to Eqs. (3.37) and (3.38), with the correlation length exponent given by Eq. (3.40) and $\eta = O(\epsilon^2)$.

One can show that the above result is completely general and not specific to the lowest order calculation nor to complex Φ . The quantum critical behavior in the above theory with $u = 0$ is identical to the classical critical behavior in the same theory in the dimension higher by one. The upper critical dimension for the transition at $T = 0$ is thus by one smaller than at $T \neq 0$.

The result is quite different when $u \neq 0$. First, in this case the coefficient v becomes irrelevant and may be set to zero. Second, to the first order in λ , only the diagram in Fig. 3.2 (b) yields a finite contribution. Setting the momenta and frequencies of the external legs to zero, the diagram in Fig. 3.2 (a), for example, gives

$$-8\lambda^2 \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{\Lambda/b}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(i\omega + q^2 - \mu)(i\omega + q^2 - \mu)}, \quad (8.12)$$

where we have also set $u = 1$. Since both poles lie in the same complex half-plane, the integral over frequencies gives zero. The diagram 3.2 (b), on the other hand, gives, at $\mu = 0$,

$$\begin{aligned} & -2\lambda^2 \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \int_{\Lambda/b}^{\Lambda} \frac{d\vec{q}}{(2\pi)^d} \frac{1}{(-i\omega + q^2)(i\omega + q^2)} \\ & = -\frac{\lambda^2 S_d \Lambda^{d-2}}{(2\pi)^d (d-2)} \left(1 - \frac{1}{b^{d-2}} \right). \end{aligned} \quad (8.13)$$

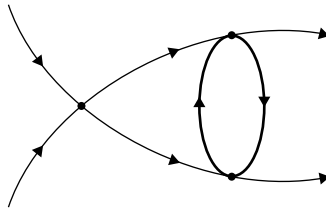
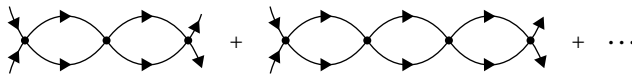
Defining a new small parameter as $\epsilon = 2 - d$, to the lowest order in ϵ the β -function for the interaction is

$$\beta_{\lambda} = \epsilon \hat{\lambda} - \hat{\lambda}^2 + O(\hat{\lambda}^3), \quad (8.14)$$

where $\hat{\lambda} = \lambda \Lambda^{d-2} S_d / (2\pi)^d$. As may already be suspected from the definition of ϵ , rescaling the momenta and the frequency now demands

$$z = 2, \quad (8.15)$$

and the upper critical dimension now equals two. The quantum critical behavior when $u \neq 0$ in both two and three dimensions is therefore the one of free bosons. This is in stark contrast with the critical behavior at finite temperatures, which as we saw lay in the universality class of the XY model. The

Figure 8.3 A vanishing third-order diagram in β_λ .Figure 8.4 Higher-order diagrams included in the β_λ in Eq. (8.14).

reason for the triviality of the quantum criticality of interacting bosons is that the normal phase is actually an empty space without any particles. The quantum critical point therefore occurs at zero density of particles and is therefore not a many-body phenomenon at all.

It is interesting to note that Eq. (8.14) is an exact result, and there are actually no higher order terms in the β_λ . The reader may convince himself that a diagram in Fig. 8.3, for example, vanishes due to one of the frequency integrals. This is in fact true for all other higher order diagrams as well. The only exception are the ones in Fig. 8.4, which are proportional to $\lambda^n (\ln b)^{n-1}$, with $n \geq 3$. Being of higher than linear order in the small quantity $\ln b$, however, these do not yield any new terms in β_λ . They are already being included in the result in Eq. (8.14), as may be realized by computing $\hat{\lambda}(b)$ from the second order β -function.

Although the critical point for $d < 2$ is evidently non-Gaussian, the quantum critical behavior at the Bose–Einstein condensation is still trivial even in $d = 1$. This is because the usual diagrams in Figs. 3.1, 3.2 (c), and 3.2 (b) that would yield the lowest-order corrections to critical exponents, together with all the higher-order diagrams, vanish due to the frequency integration. So $\nu = 1/2$, and $\eta = 0$ exactly in all dimensions, including those below the upper critical dimension. This again simply reflects that the normal phase at $T = 0$ contains no particles.

Problem 8.1 Determine the thermal and the quantum critical behavior in the model defined by the action in Eq. (8.8), assuming that the field Φ has an infinite number of complex components.

Solution Generalizing the quartic term as in Problem 2.3 and utilizing the Hubbard–Stratonovich transformation, when the number of components $N \rightarrow \infty$ the free energy becomes

$$\frac{F}{NT} = (\chi(T) - \mu)|\sigma(T)|^2 - \frac{\chi^2(T)}{2\lambda} + T \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \ln(i2u\omega_n + \omega_n^2 + k^2 + \chi(T) - \mu),$$

with $\chi(T)$ and $\sigma(T)$ satisfying the equations:

$$\frac{\chi(T)}{\lambda} = |\sigma(T)|^2 + T \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{i2u\omega_n + \omega_n^2 + k^2 + \chi(T) - \mu}$$

and

$$(\chi(T) - \mu)\sigma(T) = 0.$$

Here we set the coefficient $v = 1$, $k_B = 1$, $u \rightarrow 2u$, and assumed a uniform Hubbard–Stratonovich field $\chi(T)$ at the saddle point. Matsubara frequencies are $\omega_n = 2\pi nT$, with n being a positive or negative integer, including zero.

In the disordered phase $\sigma(T) = 0$, and we may write

$$\begin{aligned} & (\chi(T) - \mu) \\ & \times \left[1 + \lambda T \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{(i2u\omega_n + \omega_n^2 + k^2 + \chi(T) - \mu)(i2u\omega_n + \omega_n^2 + k^2)} \right] \\ & = \mu_c(T) - \mu, \end{aligned}$$

with the critical value of the chemical potential

$$\mu_c(T) = \lambda T \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{i2u\omega_n + \omega_n^2 + k^2}.$$

At $T = 0$, the sum $T \sum_{\omega_n}$ may be replaced by an integral $\int d\omega/2\pi$, and for $u = 0$ the quantum critical behavior is the same as in Problem 2.3 with one extra dimension. For $u \neq 0$, performing the frequency integral at $T = 0$, we find

$$(\chi(0) - \mu) \left[1 + \frac{\lambda}{4} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{(k^2 + u^2)^{3/2}} + O(\chi(0) - \mu) \right] = \mu_c(0) - \mu.$$

The integral over momenta is now always finite, and consequently the mean-field result

$$\chi(0) - \mu \propto \mu_c - \mu$$

obtains. At the quantum critical point $\gamma = 1$ and $\eta = 0$ in all dimensions, with the remaining critical exponent assuming their mean-field values as well.

At $T \neq 0$, on the other hand, we may separate the $\omega_n = 0$ term in the frequency sum. The rest is then always finite as $\chi(T) - \mu \rightarrow 0$, while the $\omega_n = 0$ term is finite for $d > 4$, and diverges as $(\chi(T) - \mu)^{(d-4)/2}$ for $2 < d < 4$. The thermal critical behavior in the theory is then the same as in Problem 2.3.

Finally, it is of interest to find the phase boundary in the $\mu - T$ plane. The sum over Matsubara frequencies yields

$$\mu_c(T) = \lambda \int \frac{d\vec{k}}{(2\pi)^d} \frac{n_B(u + \sqrt{u^2 + k^2}) - n_B(u - \sqrt{u^2 + k^2})}{2\sqrt{u^2 + k^2}},$$

where $n_B(z) = (e^{z/T} - 1)^{-1}$. We find:

$$\mu_c(T) - \mu_c(0) = \lambda \int \frac{d\vec{k}}{(2\pi)^d} \frac{n_B(u + \sqrt{u^2 + k^2}) + n_B(-u + \sqrt{u^2 + k^2})}{2\sqrt{u^2 + k^2}}.$$

At low temperatures $T \ll u$, therefore

$$\mu_c(T) - \mu_c(0) \propto \frac{\lambda}{2u} \int \frac{d\vec{k}}{(2\pi)^d} e^{-\frac{k^2}{2uT}} \sim T^{\frac{d}{2}},$$

in agreement with the result below Eq. (2.35). Similarly, for $u \ll T$ one finds $\mu_c(T) - \mu_c(0) \sim T^{d-1}$.

Problem 8.2 Assuming that the field Φ in the previous problem is charged, find the dependence of the magnetic field penetration depth on temperature.

Solution Let us introduce the electromagnetic potential by the usual substitution $\nabla \rightarrow \nabla - i\vec{A}(\vec{r})$, assuming that the charge is unity. Integrating out $N - 1$ components of the field, the free energy in the presence of the electromagnetic potential becomes

$$\frac{F[\vec{A}]}{NT} = \frac{F[0]}{NT} + \int d\vec{x} d\vec{y} \Pi_{ij}(\vec{x} - \vec{y}) A_i(\vec{x}) A_j(\vec{y}) + O(\vec{A}^4),$$

with the polarization

$$\Pi_{ij}(\vec{q}) = |\sigma(T)|^2 \delta_{ij} + \left(1 - \frac{1}{N}\right) \Pi'_{ij}(\vec{q}).$$

The response to the electromagnetic field consists of two parts: the first is provided by the condensate $|\sigma(T)|^2$, and the second by the remaining $N - 1$ non-condensed components. Since for $T < T_c$ the quadratic coefficient for those

$N - 1$ components is pinned at $\chi(T) - \mu = 0$, Π' is temperature independent below T_c . This guarantees that $\Pi'_{ij}(q) \rightarrow 0$ as $q \rightarrow 0$. The condensate, on the other hand, is

$$|\sigma(T)|^2 = \frac{\mu}{\lambda} - T \sum_{\omega_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{1}{i2u\omega_n + \omega_n^2 + k^2}.$$

The penetration depth is $\Lambda(T) = 1/|\sigma(T)|$, and

$$|\sigma(0)|^2 - |\sigma(T)|^2 = \frac{\mu_c(T) - \mu_c(0)}{\lambda},$$

with the right hand side as determined in the previous problem.

8.3 Bose–Hubbard model

Let us introduce the simplest model of interacting bosons that exhibits the transitions discussed in the previous section. Define the Hamiltonian

$$\hat{H}_{\text{BH}} = - \sum_{i,j} t_{ij} \hat{b}_i^\dagger \hat{b}_j - \mu \sum_{i=1}^N \hat{n}_i + \frac{U}{2} \sum_{i=1}^N \hat{n}_i(\hat{n}_i - 1), \quad (8.16)$$

where i numerates the sites of a quadratic lattice, $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$, and $\hat{b}_i^\dagger$ and $\hat{b}_i$ are the standard bosonic creation and annihilation operators satisfying the commutation relation $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$. This Hamiltonian describes a collection of bosons hopping on a lattice and interacting when on the same site with a repulsion $U > 0$. It is often called *the Bose–Hubbard model*, in analogy with its better known fermionic version.

The Bose–Hubbard model may be solved exactly when there is no hopping, and $t_{ij} \equiv 0$. Then $[\hat{H}_{\text{BH}}, \hat{n}_i] = 0$, and the ground state is simply

$$|0\rangle = \prod_i |n_i\rangle, \quad (8.17)$$

where the numbers n_i are all equal to the smallest integer larger than μ/U : $n_i = n$, for $n - 1 < \mu/U < n$. The $N(N - 1)$ lowest excited states consist simply of removing a particle at one site and placing it at another. The ground state is non-degenerate, except when μ/U is integer, when it becomes 2^N -degenerate. If the bosons were carrying an electrical charge, the system at $t_{ij} = 0$ would be an insulator, with a finite charge gap in the spectrum. It is also *incompressible*, since the particle density is, within each interval for μ , independent of the chemical potential. Since the insulating behavior is

entirely due to the interaction, such an insulator is often referred to as the *Mott insulator*.

Does the gapped insulating phase persist to finite values of hopping? With $t_{ij} \neq 0$ the Bose–Hubbard model is no longer exactly solvable, but one may still understand the qualitative features of its phase diagram using the mean-field theory first, and then considering the fluctuations.

The equilibrium behavior of the Bose–Hubbard model follows from its partition function, which, in complete analogy to Eq. (2.20), may again be expressed as a coherent-state path integral:

$$Z = \int \prod_i Db_i^*(\tau)Db_i(\tau) \exp \left(\int_0^\beta d\tau \left[\sum_i b_i^*(\tau) \partial_\tau b_i(\tau) - H_{\text{BH}}(b^*(\tau), b(\tau)) \right] \right), \quad (8.18)$$

where $H_{\text{BH}}(b^*(\tau), b(\tau))$ is given by Eq. (8.16) with $\hat{b}_i \rightarrow b_i(\tau)$, and $\hat{b}_i^\dagger \rightarrow b_i^*(\tau)$. Solvability of the interacting model at $t_{ij} = 0$ suggests that instead of the usual decoupling of the interaction term by the Hubbard–Stratonovich transformation, this time we do the same with the hopping term. This leads to the partition function expressed as an integral over the complex Hubbard–Stratonovich field $\Phi_i(\tau)$:

$$Z = \int \prod_i D\Phi_i^*(\tau)D\Phi_i(\tau) e^{-S[\Phi]}, \quad (8.19)$$

with the action

$$\begin{aligned} S[\Phi] = & \int_0^\beta d\tau \sum_{i,j} \Phi_i^*(\tau) (t^{-1})_{ij} \Phi_j(\tau) \\ & - \sum_i \ln \int Db^*(\tau)Db(\tau) e^{-\int_0^\beta d\tau L[b, \Phi_i]}. \end{aligned} \quad (8.20)$$

We are assuming that the hopping matrix t_{ij} has all the eigenvalues positive. The advantage of this strategy is that the Lagrangian,

$$\begin{aligned} L[b, \Phi_i] = & -b^*(\tau) \partial_\tau b(\tau) - \mu |b(\tau)|^2 + \frac{U}{2} |b(\tau)|^2 (|b(\tau)|^2 - 1) \\ & - \Phi_i(\tau) b^*(\tau) - \Phi_i^*(\tau) b(\tau), \end{aligned} \quad (8.21)$$

is completely local. This allows us to drop the site index on $b(\tau)$, since the functional integral in the second term in Eq. (8.20) is the same at all sites.

We may next try to expand the action in powers of Φ and its imaginary time derivatives. Such an expansion will in principle have the form

$$S[\Phi] = S[0] + \int_0^\beta d\tau \left[\sum_{i,j} \Phi_i^*(\tau)(t^{-1})_{ij} \Phi_j(\tau) + \sum_i (u \Phi_i(\tau)^* \partial_\tau \Phi_i(\tau) + v |\partial_\tau \Phi_i(\tau)|^2 + a |\Phi_i(\tau)|^2 + b |\Phi_i(\tau)|^4 + O(|\Phi_i|^6, \Phi^* \partial_\tau^3 \Phi)) \right]. \quad (8.22)$$

The coefficients u, v, a, b are in principle functions of the chemical potential, the interaction, and the temperature. They may be obtained by computing the averages of the bosonic operators over the single-site Hamiltonian. To calculate the coefficients of the terms local in imaginary time, such as a and b , however, it suffices to take a time-independent Hubbard–Stratonovich field $\Phi_i(\tau) = \Phi_i$ in $L[b, \Phi]$. In this case Φ_i enters the single-site Lagrangian in Eq. (8.21) as an external static field. In the limit $T \rightarrow 0$ the action in Eq. (8.20) then becomes

$$S[\Phi] = \sum_{i,j} \Phi_i^*(t^{-1})_{ij} \Phi_j + \frac{1}{T} \sum_i E_0[\Phi_i], \quad (8.23)$$

with $E_0[\Phi]$ as the ground state energy of the *single-site Hamiltonian*

$$\hat{H}[\Phi] = -\mu \hat{n} + \frac{U}{2} \hat{n}(\hat{n} - 1) - \Phi \hat{b}^\dagger - \Phi^* \hat{b}. \quad (8.24)$$

This follows from the recognition of the functional integral that appears in Eq. (8.20) as the single-site partition function for the above Hamiltonian.

Elementary second-order perturbation theory for the ground state energy of $\hat{H}[\Phi]$ then gives

$$E_0[\Phi] = -\left(\mu + \frac{U}{2}\right)n + \frac{U}{2}n^2 + a|\Phi|^2 + b|\Phi|^4 + O(|\Phi|^6), \quad (8.25)$$

with

$$a = \frac{n+1}{\mu - nU} + \frac{n}{-\mu + U(n-1)}. \quad (8.26)$$

The relation between the chemical potential and the number of particles at each site given below Eq. (8.17) implies then that the coefficient a is always negative. One may similarly determine the coefficient b by going to the next order in the perturbation theory. For our present purposes it will be sufficient to know that the result for b is positive for all values of the chemical potential. Finite temperature corrections to the coefficients are then exponentially small

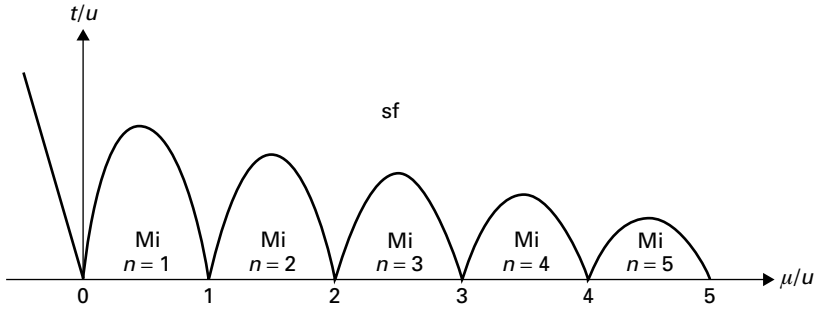


Figure 8.5 Phase diagram of the Bose–Hubbard model in the mean-field theory.

at low temperatures due to the finite gap in the spectrum of the single-site Hamiltonian $\hat{H}[\Phi]$.

For smooth configurations of $\Phi_i(\tau)$ we may take the limit of the vanishing lattice constant and finally write the continuum approximation to the action as

$$\begin{aligned}
 S[\Phi] = S[0] + \int_0^\beta d\tau \int d\vec{x} & \left[u \Phi^*(\vec{x}, \tau) \partial_\tau \Phi(\vec{x}, \tau) + v |\partial_\tau \Phi(\vec{x}, \tau)|^2 \right. \\
 & + w |\nabla \Phi(\vec{x}, \tau)|^2 + \tilde{a} |\Phi(\vec{x}, \tau)|^2 + b |\Phi(\vec{x}, \tau)|^4 \\
 & \left. + O(|\Phi|^6, \Phi^* \partial_\tau^3 \Phi, |\nabla^2 \Phi|^2) \right], \quad (8.27)
 \end{aligned}$$

where we have also employed the usual gradient expansion. Defining $t(\vec{q})$ as the Fourier transform of the hopping matrix, the coefficient $\tilde{a}$ becomes

$$\tilde{a} = \frac{1}{t} + a, \quad (8.28)$$

where $t = t(\vec{q} = 0)$. Neglecting the fluctuations, $\langle \Phi \rangle \neq 0$ for $\tilde{a} < 0$, or, when the hopping is larger than the critical value,

$$\frac{t_c}{U} = \frac{(n - \frac{\mu}{U})(\frac{\mu}{U} - n + 1)}{\frac{\mu}{U} + 1}. \quad (8.29)$$

The reader can check that the average $\langle \Phi(x, \tau) \rangle \sim \langle b(x, \tau) \rangle$, so that the average value of the Hubbard–Stratonovich field is proportional to the superfluid order parameter. For $t > t_c$ the system is therefore in the superfluid phase, which is gapless and compressible. Below t_c the system is the Mott insulator with the gap. t_c therefore determines the mean-field superfluid–Mott insulator phase boundary at which the Mott gap collapses and superfluid order develops. It has the shape as shown in Fig. 8.5. Note that t_c vanishes at μ/U integer. Since at these special values of μ the ground state energy at $t = 0$ is macroscopically degenerate, an infinitesimal hopping is enough to select

the superfluid ground state. Otherwise $t_c > 0$. There is therefore always a maximal t_c at a given n which is located near and below $\mu/U = n - (1/2)$.

So far we have confined the discussion to the determination of the coefficients of the terms local in imaginary time in the expansion in Eq. (8.22). This, however, turns out to be enough to fix all the remaining coefficients. Once the coefficient a has been found, for example, the coefficients u and v follow from the “gauge symmetry” of the action in Eq. (8.20). Notice that the global time-dependent transformation

$$b(\tau) \rightarrow b(\tau)e^{i\theta(\tau)}, \quad (8.30)$$

$$b^*(\tau) \rightarrow b^*(\tau)e^{-i\theta(\tau)}, \quad (8.31)$$

$$\Phi_i(\tau) \rightarrow \Phi_i(\tau)e^{i\theta(\tau)}, \quad (8.32)$$

$$\Phi_i^*(\tau) \rightarrow \Phi_i^*(\tau)e^{-i\theta(\tau)}, \quad (8.33)$$

$$\mu \rightarrow \mu + i\partial_\tau\theta(\tau), \quad (8.34)$$

leaves the form of the action invariant. Although the chemical potential under the gauge transformation becomes complex, this property may be exploited to find the coefficients u and v , for example. Demanding that the action in Eq. (8.27) is also gauge-symmetric after some algebra implies

$$\frac{\partial \tilde{a}}{\partial \mu} + u = 0, \quad (8.35)$$

$$\frac{\partial u}{\partial \mu} - 2v = 0, \quad (8.36)$$

and similarly for the coefficients of higher-order time derivatives. These equations are completely general. Using our particular result for $\tilde{a}$ yields

$$u = -\frac{1}{t_c^2} \frac{\partial t_c}{\partial \mu}, \quad (8.37)$$

and

$$v = \frac{1}{t_c^3} \left(\frac{\partial t_c}{\partial \mu} \right)^2 - \frac{1}{2t_c^2} \frac{\partial^2 t_c}{\partial \mu^2}, \quad (8.38)$$

with the critical hopping as the function of chemical potential as in Eq. (8.29). Note that the coefficient v is thus always positive, while u changes sign at the value of the chemical potential at which t_c is maximal. So whenever the superfluid transition may be tuned by changing the chemical potential, it is in the universality class of the $T = 0$ Bose–Einstein condensation. At the tip of the insulating lobe, on the other hand, the superfluid transition can be induced

only by increasing hopping, and the mean-field theory suggests that it then lies in the universality class of the XY model in $d + 1$ dimensions.

The principal features of the phase diagram are in fact correct beyond the mean-field theory we used. First, since the Mott insulator has a gap in the spectrum at $t = 0$, it must take a finite hopping to collapse it, except at integer values of μ/U when the gap vanishes. Since the collapse of the gap coincides with the emergence of the superfluid order in the system, the exact superfluid–insulator phase boundary has the shape qualitatively similar to the mean-field result. Second, changing hopping inside the insulating phase cannot change the density of particles. Since the hopping part of the Hamiltonian commutes with total particle number, the ground state at small $t \neq 0$ is still an eigenstate of the total number of particles. Since there is a finite gap, on the other hand, the total number of particles in the ground state at small $t \neq 0$ and $t = 0$ can differ only by an amount proportional to t/U . Discreteness of the particle number then implies that it remains constant as long there is a gap in the spectrum. Since the particle density within a lobe thus remains the same along the line $t = 0$, and cannot change by increasing hopping as long as there is a gap in the spectrum, the whole insulating lobe corresponds to the same integer particle number per site.

The nature of the superfluid–insulator quantum phase transition may be inferred as follows. Since the gauge invariance is an exact symmetry in the theory, the general Eqs. (8.35) and (8.36) must continue to hold between the renormalized coefficients in the low-energy action, which would be obtained by integrating out the high-energy fluctuations of the order parameter. The transition is then still determined by the condition $\tilde{a}(\mu, t) = 0$, but with the exact $\tilde{a}(\mu, t)$ being fully renormalized by the fluctuations. It is easy to see that the coefficient u then still has to vanish at the tip of the insulating lobe. If we consider the two-dimensional gradient of the function $\tilde{a}(\mu, t)$, at the tip of any line of constant $\tilde{a}(\mu, t)$ the gradient will become orthogonal to the μ -axis, so that $\partial\tilde{a}(\mu, t)/\partial\mu = 0$ at the line of such tips. Gauge invariance of the action implies then that at such a line of tips the coefficient $u = 0$. So there exists a line in the μ – t plane that goes through the tip of the insulating lobe along which the superfluid–insulator transition is indeed in the XY universality class. In the mean field theory this line is approximated by a straight line parallel to the hopping axis. Away from this line the transition is in the universality class of the $T = 0$ Bose–Einstein condensation.

It is useful to consider the line of constant integer density in the superfluid phase. First, this line also must meet the insulating lobe precisely at the tip: assuming that it does not would lead to a negative compressibility in a finite

region near the tip, which is physically impossible. Second, the density of bosons as a function of the chemical potential and the hopping, $N(\mu, t)$, being a fixed integer everywhere inside the insulating lobe, is such that the sign of $\partial N(\mu, t)/\partial t$ at $t = t_c + 0$ changes from negative to positive at the phase boundary as one goes around the lobe in the direction of increasing chemical potential. It therefore has to vanish somewhere at the phase boundary. At that point increasing hopping by an infinitesimal amount does not change the integer density of particles. Since there could be only one such line with the integer density ending at the given insulating lobe, and that right at the tip, at the tip $\partial N(\mu, t)/\partial t = 0$. The line of integer density meets the tip orthogonally to the axis of the chemical potential. One can show explicitly that this indeed is what happens within the mean-field theory. The line of integer density and the line $u = 0$, however, are in principle different.

Problem 8.3 Show that the Bose–Hubbard model has the superfluid ground state at integer μ/U for any value of hopping by using the degenerate perturbation theory.

Solution When $\mu/U = n$ and $\Phi = 0$, the single-site Hamiltonian $\hat{H}[0]$ in Eq. (8.24) has two degenerate ground states, $|n\rangle$ and $|n+1\rangle$. In this basis $\hat{H}[\Phi]$ is represented by a two by two matrix with $-nU(n+1)/2$ on the diagonal, and $-\sqrt{n+1}\Phi$ and its complex conjugate as off-diagonal elements. The ground state energy is therefore

$$E_0[\Phi] = -\frac{U}{2}n(n+1) - (n+1)|\Phi| + O(|\Phi|^2).$$

Since the positive hopping term in $S[\Phi]$ in Eq. (8.23) is quadratic in Φ , the value of the action can always be decreased by developing a weak superfluid order.

Problem 8.4* Compute the coefficients u and v in Eq. (8.27) without the use of the gauge symmetry.

Solution Expanding the exponential in Eq. (8.20) in powers of Φ to quadratic order yields

$$\begin{aligned} S[\Phi(\tau)] = & S[0] + \int_0^\beta d\tau \sum_{i,j} \Phi_i^*(\tau)(t^{-1})_{ij} \Phi_j(\tau) \\ & - \sum_i \int \frac{d\omega}{2\pi} \langle b^*(\omega)b(\omega) \rangle |\Phi_i(\omega)|^2, \end{aligned}$$

where the average is supposed to be taken over the Lagrangian in Eq. (8.21) with $\Phi_i = 0$. The average is

$$\langle b^*(\omega)b(\omega) \rangle = \int_{-\infty}^{\infty} d(\tau_2 - \tau_1) e^{-i\omega(\tau_2 - \tau_1)} \langle n | T_{\tau} \{ \hat{b}(\tau_1), \hat{b}^{\dagger}(\tau_2) \} | n \rangle,$$

where $|n\rangle$ is the ground state of the single-site Hamiltonian $\hat{H}[0]$ in Eq. (8.24), and $\hat{b}(\tau) = e^{\hat{H}[0]\tau} \hat{b} e^{-\hat{H}[0]\tau}$, $\hat{b}^{\dagger}(\tau) = e^{\hat{H}[0]\tau} \hat{b}^{\dagger} e^{-\hat{H}[0]\tau}$. T_{τ} is the time-ordering operator that places operators at earlier times to the right of those at later times. So,

$$\begin{aligned} \langle b^*(\omega)b(\omega) \rangle &= \int_0^{\infty} d\tau e^{-i\omega\tau} \langle n | \hat{b}^{\dagger}(\tau) \hat{b}(0) | n \rangle + \int_0^{\infty} d\tau e^{i\omega\tau} \langle n | \hat{b}(\tau) \hat{b}^{\dagger}(0) | n \rangle \\ &= \frac{n}{i\omega + \mu - U(n-1)} + \frac{n+1}{-i\omega - \mu + nU}. \end{aligned}$$

At $\omega = 0$ this reproduces the mean-field expression for the coefficient a in Eq. (8.26). The Taylor expansion in powers of frequency yields the coefficients u and v , in accordance with the gauge symmetry.

8.4 Quantum fluctuations and the superfluid density

It is sometimes convenient to express the complex functions $b_i(\tau)$, $b_i^*(\tau)$ in the partition function of the Bose–Hubbard model in Eq. (8.18) in terms of their amplitude and the phase:

$$b_i(\tau) = \rho_i^{1/2}(\tau) e^{i\phi_i(\tau)}. \quad (8.39)$$

In the superfluid phase the variable $\rho_i(\tau)$ can be viewed as the fluctuating local density of particles. Writing $\rho_i(\tau) = \rho_0 + \delta\rho_i(\tau)$, with $\rho_0 = \mu/U + 1/2$ as the average particle number per site, and expanding the action in Eq. (8.18), one finds $S = \int_0^{\beta} d\tau L$ with the Lagrangian

$$\begin{aligned} L &= i(\rho_0 + \delta\rho_i(\tau)) \sum_i \dot{\phi}_i(\tau) \\ &\quad - \rho_0 \sum_{\langle ij \rangle} t_{ij} \cos(\phi_i(\tau) - \phi_j(\tau)) + \frac{U}{2} \sum_i \delta\rho_i^2(\tau). \end{aligned} \quad (8.40)$$

Here we assumed nearest-neighbor hopping and neglected the imaginary time derivative of the density, as well as deviations from the average density in the hopping term. The latter approximation is justified if $\rho_0 \gg 1$.

The imaginary time in the Lagrangian that couples local density to the time derivative of the local phase is reminiscent of the term in the Feynman's path integral for a quantum mechanical particle, with the density playing the role of

the momentum, and the phase of the position observable. By analogy, then, the density and the phase may be considered as canonically conjugate variables, satisfying an analogous commutation relation. This provides an alternative view at the superfluid–insulator transition from the last section. In the insulating phase the quantum uncertainty of the local density of particles is small. The uncertainty relation implies then that the fluctuation of the canonically conjugate phase variable must be large. Similarly to thermal fluctuations at finite temperature, these *quantum fluctuations* of the phase may be enough to destroy the long-range order in the system. In the superfluid phase, on the other hand, the situation becomes reversed.

Quantum fluctuations become most apparent if we integrate over the small density deviations $\delta\rho_i$ in L . In terms of the remaining phase variable the Lagrangian then reduces to

$$L = i\rho_0 \sum_i \dot{\phi}_i(\tau) - \rho_0 \sum_{\langle ij \rangle} t_{ij} \cos(\phi_i(\tau) - \phi_j(\tau)) + \frac{1}{2U} \sum_i \dot{\phi}_i^2(\tau). \quad (8.41)$$

When $U = 0$ the fluctuations of $\phi_i(\tau)$ in imaginary time become completely suppressed, and the partition function in the case of nearest-neighbor hopping reduced to the XY model in Eq. (6.29), with $J = t\rho_0$. When U is large, on the other hand, imaginary time fluctuations of the phase become significant. Furthermore, once the fluctuations in imaginary time become allowed the Lagrangian is no longer always real. Since the periodic boundary conditions for $b_i(\tau)$ imply that $\phi_i(\beta) = \phi_i(0) + 2\pi n_i$, the action is real only at an integer density. Note that, for the assumed large ρ_0 , this happens at the tip of the insulating lobe, located at $\mu/U = n - 1/2$ for large n .

One may further understand the role of quantum fluctuations by computing the superfluid density, i.e. the stiffness of the phase, using the variational method in which the action defined by the above Lagrangian is replaced by the optimal quadratic one (Problem 6.4):

$$S_0 = \frac{T}{2} \sum_{\omega_n, \vec{k}} \phi(\omega_n, \vec{k}) G_0^{-1}(\omega_n, \vec{k}) \phi(-\omega_n, -\vec{k}). \quad (8.42)$$

S_0 is to be chosen to minimize the variational free energy $F_{\text{var}} = F_0 + T \langle S - S_0 \rangle_0$, with the last average taken over the ensemble defined by S_0 . At low temperatures only small fluctuations in imaginary time are important, so that all the finite winding numbers $n_i \neq 0$ may be neglected. We may therefore drop the imaginary term in the action. The variational free energy is then

easily computed:

$$F_{\text{var}} = -\frac{T}{2} \sum_{\omega_n, \vec{k}} \ln G_0(\omega_n, \vec{k}) + \frac{T}{2U} \sum_{\omega_n, \vec{k}} \omega_n^2 G_0(\omega_n, \vec{k}) - J \sum_{\hat{r}} e^{-T \sum_{\omega_n, \vec{k}} G_0(\omega_n, \vec{k}) (1 - \cos(\vec{k} \cdot \hat{r}))}, \quad (8.43)$$

where $\hat{r}$ is a lattice unit vector, and we assumed the nearest-neighbor hopping matrix t_{ij} . Minimizing with respect to $G_0(\omega_n, \vec{k})$ then yields

$$G_0^{-1}(\omega_n, \vec{k}) = \frac{\omega_n^2}{U} + 4K \sum_{\hat{r}} \sin^2 \left(\frac{\vec{k} \cdot \hat{r}}{2} \right), \quad (8.44)$$

with K as

$$K = J \exp \left(-\frac{1}{2d\sqrt{K}} \int \frac{d\vec{q}}{(2\pi)^2} f(\vec{q}) \coth \left(\frac{\sqrt{K}}{T} f(\vec{q}) \right) \right), \quad (8.45)$$

with

$$f^2(\vec{k}) = U \sum_{\hat{r}} \sin^2 \left(\frac{\vec{k} \cdot \hat{r}}{2} \right). \quad (8.46)$$

The temperature-dependent quantity K evidently is the coefficient of the $(\nabla\phi)^2$ term in S_0 , and should therefore be identified with the phase stiffness.

At $T = 0$ the superfluid density is reduced by the quantum fluctuations as

$$\frac{K(0)}{J} = 1 - \frac{1}{2dJ} \int \frac{d\vec{q}}{(2\pi)^2} f(\vec{q}) + O(U). \quad (8.47)$$

At $U = 0$, on the other hand, K in the XY model at low temperatures satisfies

$$K = J e^{-\frac{T}{2dK}} \approx J - \frac{T}{2d} + O(T^2). \quad (8.48)$$

The last result agrees well with Monte Carlo calculations, including the coefficient of the term linear in temperature. Finally, for finite U and at temperatures well below the temperature scale for quantum fluctuations $\omega_q^{2(d+1)} = U^d K(0)^{d+2}$,

$$K \approx K(0) \left(1 - I(d) \left(\frac{T}{\omega_q} \right)^{d+1} \right), \quad (8.49)$$

with the coefficient $I(d)$ given by the integral

$$I(d) = \frac{1}{2d} \int \frac{d\vec{q}}{(2\pi)^d} q e^{-q}. \quad (8.50)$$

The interactions thus have a twofold effect. First, by introducing phase fluctuations in the direction of the imaginary time they reduce the stiffness of the phase even at $T = 0$. Second, they suppress further reduction of the stiffness with temperature by introducing the quantum energy scale ω_q below which the power in the power-law behavior of phase stiffness vs. temperature becomes higher than one. The $\sim T^4$ dependence in three dimensions is in excellent agreement with the measurements of the superfluid density in ^4He .

Problem 8.5 Estimate the critical temperature of the XY model by using the variational result for the superfluid density.

Solution Using the universal value of the superfluid density jump at the Kosterlitz–Thouless transition, $K(T_c)/T_c = 2/\pi$ in Eq. (6.80) gives

$$\frac{T_c}{J} = \frac{\pi}{2} e^{-\frac{\pi}{8}} = 1.06.$$

Monte Carlo simulations give an essentially exact result, $T_c/J = 0.90$, for the two-dimensional XY model. The variational approximation overestimates the critical temperature primarily due to its neglect of vortices, which are very efficient in reducing the phase stiffness at temperatures near T_c .

Problem 8.6 Find the reduction of the superfluid density at low temperatures in two and three dimensions in the case of Coulomb interaction.

Solution Generalizing to an arbitrary interaction, the variational approximation gives F_{var} as in Eq. (8.43) with $1/U$ replaced with $1/V(\vec{k})$ under the momentum sum. This yields the stiffness K being determined by the same Eq. (8.45), except that

$$f^2(\vec{k}) = V(k) \sum_{\hat{r}} \sin^2 \left(\frac{\vec{k} \cdot \hat{r}}{2} \right).$$

In two dimensions $V(k) = e^2/|k|$ at small momenta, and

$$\frac{K(T)}{K(0)} = 1 - \frac{6}{\pi} \left(\frac{T}{\omega_q} \right)^5.$$

with $\omega_q = e^{4/5} K^{3/5}(0)$. Similarly, in three dimensions $V(k) = e^2/k^2$ so that

$$K(0) - K(T) \propto \omega_q e^{-\frac{\omega_q}{T}},$$

with $\omega_q^2 = e^2 K_0$ being the plasma frequency.

8.5 Universal conductivity in two dimensions*

If we consider the bosons in the Bose–Hubbard model as possessing an electrical charge e^* , further interesting results for the conductivity of the system in two dimensions may be deduced. If we are interested only in the conductivity near the superfluid–insulator transition, we may compute it as a response of the system to a weak external vector potential $\vec{a}(\vec{r}, \tau)$, with $\vec{a}(\vec{r}, \tau)$ entering the partition function via the usual replacement

$$\nabla \rightarrow \nabla - i \frac{e^*}{\hbar} \vec{a}(\vec{r}, \tau) \quad (8.51)$$

in the action for the superfluid order parameter in Eq. (8.27). The equality of the charge of the Hubbard–Stratonovich field Φ and the original bosons follows from the gauge invariance of the action in Eq. (8.20). The average electrical current is then

$$\langle j_i(\vec{r}, \tau) \rangle = -\hbar \frac{\delta F}{\delta a_i(\vec{r}, \tau)}, \quad (8.52)$$

where $F = -\ln Z$, and Z is the partition function. The conductivity tensor σ_{ij} may then be defined at imaginary times as determining the response to an external electric field,

$$\langle j_i(\vec{r}, \tau) \rangle = \int d\vec{r}' d\tau' \sigma_{ij}(\vec{r} - \vec{r}', \tau - \tau') E_j(\vec{r}', \tau'), \quad (8.53)$$

where the electric field is

$$E_i(\vec{r}, \tau) = -i \frac{\partial a_i(\vec{r}, \tau)}{\partial \tau}. \quad (8.54)$$

The last equation becomes the equivalent of the standard relation between the electric field and the electromagnetic vector potential in real time t upon substitution $\tau = it$.

Performing a partial integration in Eq. (8.53) and then differentiating with respect to the vector potential once again, we find

$$\frac{1}{i} \frac{\partial \sigma_{ij}(\vec{r} - \vec{r}', \tau - \tau')}{\partial \tau'} = \hbar \frac{\delta^2 F}{\delta a_j(\vec{r}', \tau') \delta a_i(\vec{r}, \tau)}. \quad (8.55)$$

Evaluation of the derivative on the right hand side at $\vec{a}(\vec{r}, \tau) \equiv 0$ gives then the linear conductivity. Its uniform ($\vec{q} = 0$) component is then

$$\sigma_{ij}(q = 0, \omega) = \frac{\rho_{ij}(\omega)}{\omega} \frac{e^{*2}}{\hbar}, \quad (8.56)$$

with the function $\rho_{ij}(\omega)$ as

$$\rho_{ij}(\omega) = 2w\delta_{ij}\langle\Phi^*(0,0)\Phi(0,0)\rangle - w^2 \int d\vec{r} \int_0^\beta d\tau e^{i\omega\tau} \langle j_i(\vec{r}, \tau) j_j(0,0) \rangle \quad (8.57)$$

and the current $\vec{j}(\vec{r}, \tau) = i\Phi^*(\vec{r}, \tau)\nabla\Phi(\vec{r}, \tau) - i\Phi(\vec{r}, \tau)\nabla\Phi^*(\vec{r}, \tau)$. The last result is an example of the general Kubo formula for the linear response to an external perturbation. The two terms appearing in $\rho_{ij}(\omega)$ are often called the “diamagnetic” and the “paramagnetic” contributions, respectively. The response in real physical time may be obtained from the knowledge of $\sigma_{ij}(\omega)$ by performing the analytical continuation from Matsubara to real frequencies in the last equation, as $i\omega \rightarrow \omega + i\eta$.

The form of the conductivity near the transition may be easily derived by noticing that Eq. (8.57) implies that it scales with length as $\sigma \sim L^{d+2z}j^2$. Scaling of the electrical current j , on the other hand, may be determined exactly, since the gauge invariance dictates that the form $\nabla - (e/\hbar)\vec{a}$ remains preserved during the momentum shell transformation, so that $\vec{a} \sim L^{-1}$. Since $jaL^{d+z} \sim 1$, we find that $\sigma \sim L^{2-d}$. In an isotropic system $\sigma_{ij}(q=0, \omega) = \sigma(\omega)\delta_{ij}$, and we can write

$$\sigma(\omega) = \xi^{2-d} F_{\pm} \left(\frac{\hbar\omega}{k_B T}, \frac{\hbar\omega_0|\delta|^{z\nu}}{k_B T} \right) \frac{(e^*)^2}{\hbar}, \quad (8.58)$$

where $\xi \sim 1/|\delta|^\nu$ is the correlation length, $\hbar\omega_0$ is the characteristic microscopic energy scale, and δ the dimensionless tuning parameter for the superfluid–insulator transition. $F_{\pm}$ are two universal dimensionless functions of two dimensionless variables, corresponding to the two sides of the transition.

Of course, the scaling form for the conductivity is just a special case of the general scaling relations near critical points. What is remarkable about it, however, is that we were able to determine the dimensionality of conductivity, namely the power $2-d$, exactly. The anomalous dimension of the current vanishes, and its scaling with length is determined exactly by its engineering dimension. This is a consequence of the gauge-invariant way the vector potential enters the action. Another example of the same phenomenon was provided by the exact result for the anomalous dimension of the gauge field in the Ginzburg–Landau theory in Section 4.4.

The scaling relation in Eq. (8.58) has an interesting and a rather non-intuitive physical consequence. In $d=2$ the exponent of the correlation length

dependent prefactor vanishes, and at the critical point at $\delta = 0$,

$$\sigma_c(\omega) = f\left(\frac{\hbar\omega}{k_B T}\right) \frac{(e^*)^2}{\hbar}, \quad (8.59)$$

where $f(x) = F_{\pm}(x, 0)$ is a universal function. In particular, the dc conductivity at the superfluid transition is a universal number in units of $(e^*)^2/\hbar$. If this number is finite, this implies that at $T = 0$, and right at the critical point between insulating and superfluid phases the system becomes a curious metal, the conductivity of which is a constant of nature. Something resembling this phenomenon is observed in thin metallic films with thickness used as a tuning parameter. Assuming that near the quantum critical point Cooper pairs are the effective bosons in the problem, the charge e^* is twice the electron charge, and $\hbar/e^{*2} \approx 1000$ ohm. The experiment suggest then that $f(0) \approx 0.8 - 0.9$. This number, however, is not directly relevant to the Bose–Hubbard model, since the long-range Coulomb interaction and disorder, both present in real experiments, change the universality class of the superfluid–insulator transition.

The difficulties of the computation of the function $f(x)$ may be appreciated by considering the two opposite limits of $\hbar\omega/k_B T \ll 1$ and $k_B T/\hbar\omega \ll 1$, and by realizing that they are determined by two entirely different mechanisms of dissipation. The former corresponds to the transport of pre-existing thermally excited quasiparticles, which get accelerated by the electric field, and relax back to equilibrium via mutual collisions. The system is therefore in the hydrodynamic regime, and the transport is in principle described by the appropriate Boltzmann equation, similarly to the Drude conductivity of a metal. It is indeed quite remarkable that the conductivity in this limit of completely incoherent transport is nevertheless universal. The latter limit, on the other hand, corresponds to the collisionless transport of coherent excitations created by the electric field. The two limits of $f(x)$ therefore may be expected to be different, and indeed calculations show that $f(0) > f(\infty)$. It is important to realize that both limits in principle describe the transport in the system at low temperatures. The experiments, however, typically lie in the hydrodynamic regime where $\hbar\omega/k_B T \ll 1$, and the measurements should in principle yield the value of $f(0)$.

Computation of $f(x)$ may be performed using the $\epsilon = 3 - d$ expansion, or the large- N expansion in $d = 2$. It becomes, however, quite technical, and we will describe here only the evaluation of the leading term in ϵ -expansion, which will be sufficient to get the correct physical picture. First, assume that the electromagnetic potential, and therefore the current, is along the x -direction. Neglecting the interaction term in the action in Eq. (8.27), at finite

temperatures we can write

$$\rho(\omega) = 2c^2 k_B T \sum_{v_n} \int \frac{d\vec{k}}{(2\pi)^d} \left(\frac{1}{v_n^2 + c^2 k^2 + m^2} - \frac{2c^2 k_x^2}{(v_n^2 + c^2 k^2 + m^2)((v_n + \omega)^2 + c^2 k^2 + m^2)} \right), \quad (8.60)$$

where $c^2 = w/v$, $m^2 = \tilde{a}/v$, and we assumed $u = 0$ for convenience. The last equation therefore gives the conductivity at the superfluid–insulator transition in the XY universality class, to the leading order in small $\epsilon = 3 - d$. The expression may be simplified by inserting the unity under the first integral written as $\partial k_x / \partial k_x = 1$, and then integrating by parts. Since the boundary term vanishes, we may write

$$\rho(\omega) = \sum_{v_n} \int \frac{d\vec{k}}{(2\pi)^d} \frac{4c^4 k_B T k_x^2}{v_n^2 + c^2 k^2 + m^2} \times \left(\frac{1}{v_n^2 + c^2 k^2 + m^2} - \frac{1}{(v_n + \omega)^2 + c^2 k^2 + m^2} \right). \quad (8.61)$$

Written in this form, it becomes evident that $\rho(\omega)$ vanishes at small Matsubara frequency. Performing the sums over the Matsubara frequencies, we find $\sigma(\omega) = \sigma_1(\omega) + \sigma_2(\omega)$, with

$$\sigma_1(\omega) = \frac{2c^{2-d}}{\omega} \int \frac{d\vec{k}}{(2\pi)^d} \frac{k_x^2}{e_k^2} \left(-\frac{\partial n_B(e_k)}{\partial e_k} \right) \frac{e^{*2}}{\hbar}, \quad (8.62)$$

$$\sigma_2(\omega) = \omega c^{2-d} \int \frac{d\vec{k}}{(2\pi)^d} \frac{k_x^2}{e_k^3 (\omega^2 + 4e_k^2)} (1 + 2n_B(e_k)) \frac{e^{*2}}{\hbar}, \quad (8.63)$$

and $e_k^2 = k^2 + m^2$.

The two contributions to conductivity have very different physical origins. Analytically continuing to real frequencies and using the identity

$$\lim_{\eta \rightarrow 0} \frac{1}{\omega + i\eta} = P \left(\frac{1}{\omega} \right) - i\pi \delta(\omega), \quad (8.64)$$

in $d = 3$ we obtain

$$\text{Re} \sigma_1(\omega) = \frac{(k_B T)^2}{3\pi c} \delta(\omega) f_1 \left(\frac{m}{k_B T} \right) \frac{(e^*)^2}{\hbar}, \quad (8.65)$$

with

$$f_1(z) = \int_0^\infty dt \frac{t^4}{t^2 + z^2} \frac{e^{\sqrt{z^2 + t^2}}}{(e^{\sqrt{z^2 + t^2}} - 1)^2} = \frac{\pi^2}{3} + O(z). \quad (8.66)$$

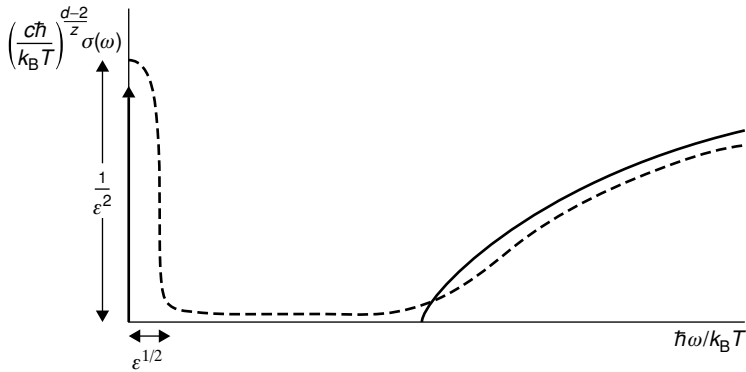


Figure 8.6 Conductivity in the insulating phase: the full line represents the result in the absence of interactions, with the heavy arrow at $\omega = 0$ as the delta-function. The dashed line is the result to second order in $\epsilon = 3 - d$.

For $m \ll k_B T$ and in $d = 3$, therefore,

$$\text{Re}\sigma_1(\omega) = \frac{\pi}{9} \left(\frac{k_B T}{c\hbar} \right) \delta \left(\frac{\hbar\omega}{k_B T} \right) \frac{e^{*2}}{\hbar}. \quad (8.67)$$

Similarly, analytically continuing to real frequencies one finds

$$\text{Re}\sigma_2(\omega) = \frac{1}{48\pi} \frac{\omega}{c} \left(\frac{\omega^2 - 4m^2}{\omega^2} \right)^{3/2} \left(1 + 2n_B \left(\frac{\hbar\omega}{2} \right) \right) \Theta(\omega - 2m). \quad (8.68)$$

Both terms indeed acquire the announced scaling form upon identification of the correlation length as $\xi = 1/m$, with the dynamical exponent $z = 1$. σ_2 vanishes below the threshold for the creation of a particle–hole pair, and describes the collisionless transport of coherent field-induced quasiparticles. σ_1 , on the other hand, albeit vanishing at zero temperature, becomes proportional to $\delta(\omega)$ at finite temperatures. This part in principle describes the collision-dominated transport of incoherent, thermally generated excitations. In the Gaussian approximation, however, the excitations are completely coherent, and yield therefore the response of an ideal conductor. Including interactions will provide them with finite lifetimes, and one expects that the delta-function at $\omega = 0$ will be spread into a Lorentzian, as in Fig. 8.6. This is indeed what is found to the second order in interactions. The universality of the collision-dominated transport derives from the fact that the interaction itself becomes a universal number of order ϵ near the quantum critical point.

Problem 8.7* Compute the universal conductivity at the superfluid–insulator transition in two dimensions, in the regime $\hbar\omega/k_B T \gg 1$, in Gaussian approximation.

Solution In the collisionless limit we may set $T = 0$ in Eq. (8.63). At the critical point $m = 0$, and the straightforward integration gives $\sigma_2 = (1/16)(e^{*2}/\hbar)$.

Problem 8.8* Determine the lowest order correction in $\epsilon = 3 - d$ to the weight of the delta-function peak in $\sigma_1(\omega)$, at a temperature T right above the quantum critical point.

Solution First, let us determine the shift in the critical point at finite temperature. To the lowest order in interaction, we find

$$m^2(T) = -\mu + 4\lambda T \sum_{\omega_n} \frac{1}{\omega_n^2 + k^2 - \mu} + O(\lambda^2),$$

where we set $u = 0$, $v = 1$ in the action in Eq. (8.8). Adding and subtracting the $T = 0$ contribution, we may write

$$m^2(T) = m^2(0) + 4\lambda \int \frac{d\vec{k}}{(2\pi)^d} \frac{n_B(\sqrt{k^2 + m^2(0)})}{\sqrt{k^2 + m^2(0)}} + O(\lambda^2).$$

At the quantum critical point at $m(0) = 0$, therefore,

$$m^2(T) = \frac{4\lambda S_d}{(2\pi)^d} T^{d-1} \int_0^\infty \frac{k^{d-2} dk}{e^k - 1}.$$

To the lowest order in $\epsilon = 3 - d$ we need to evaluate the above integral at $d = 3$, and replace $\hat{\lambda}$ as defined below Eq. (8.11) with its fixed point value, $\hat{\lambda} = \epsilon/10$. At the critical point, therefore,

$$\frac{m(T)}{T} = \frac{2\pi}{\sqrt{15}} \epsilon^{1/2} + O(\epsilon^{3/2}).$$

Expanding, on the other hand, the function $f_1(z)$ in Eq. (8.66) around $z = 0$ yields

$$f_1(z) = \frac{\pi^2}{3} \left(1 - \frac{9}{4\pi} z + O(z^2) \right),$$

so that

$$\text{Re}\sigma_1(\omega) = \frac{\pi}{9} \left(1 - \frac{9}{2\sqrt{15}} \sqrt{\epsilon} + O(\epsilon) \right) \left(\frac{k_B T}{c\hbar} \right) \delta \left(\frac{\hbar\omega}{k_B T} \right) \frac{e^{*2}}{\hbar}.$$

Problem 8.9* Determine the relevance of Coulomb interaction between bosons at the quantum transition to a superfluid state in two dimensions.

Solution Let us add the term

$$\int_0^{1/T} d\tau \int d\vec{x} d\vec{y} |\Phi(\vec{x}, \tau)|^2 \frac{e^2}{2|\vec{x} - \vec{y}|} |\Phi(\vec{y}, \tau)|^2$$

to the action for interacting bosons in Eq. (8.8), with $u = 1$ and $v = 0$. Introducing the Hubbard–Stratonovich scalar gauge field $a(\vec{x}, \tau)$ to decouple the Coulomb interaction, we may write the action at $T = 0$ as

$$S = \int d\tau d\vec{x} [\Phi^*(\vec{x}, \tau)(\partial_\tau - ia(\vec{x}, \tau))\Phi(\vec{x}, \tau) + |\nabla\Phi(\vec{x}, \tau)|^2 - \mu|\Phi(\vec{x}, \tau)|^2 + \lambda|\Phi(\vec{x}, \tau)|^4 + \frac{1}{2e^2}a(\vec{x}, \tau)|\nabla a(\vec{x}, \tau)|^2],$$

with the operator $|\nabla|$ to be understood as $|\vec{q}|$ in Fourier space. The neutralizing background can be included by omitting the $\vec{k} = 0$ mode for the gauge field. In this form the first two terms in the action are invariant under a limited gauge transformation

$$\begin{aligned}\Phi(\vec{x}, \tau) &\rightarrow e^{i\phi(\tau)}\Phi(\vec{x}, \tau), \\ a(\vec{x}, \tau) &\rightarrow a(\vec{x}, \tau) + \partial_\tau\phi(\tau).\end{aligned}$$

This guarantees that their form remains invariant under the momentum-shell transformation. So the gauge field scales exactly as $a \sim L^{-z}$ with length, which then implies $e^2 \sim L^{1-z}$.

The correction to the order parameter quadratic action still vanishes. Also, since the quadratic term for the gauge field is non-analytic in momentum, it cannot renormalize. This implies that the β -functions for the two interaction couplings are

$$\frac{d\hat{e}^2}{d\ln b} = (z - 1)\hat{e}^2,$$

with $\hat{e}^2 = e^2\Lambda^{1-z}$ and with $z = 2$, and Eq. (8.14) with $\epsilon = 0$. Whereas the short-range interaction is marginally irrelevant in two dimensions, a weak Coulomb interaction is relevant. This should be interpreted as that the system suffers a first-order transition from the superfluid state into the Wigner crystal as the density of bosons is decreased.

Problem 8.10* Determine the universality class of the superfluid transition in the system of lattice bosons at a commensurate density and in general

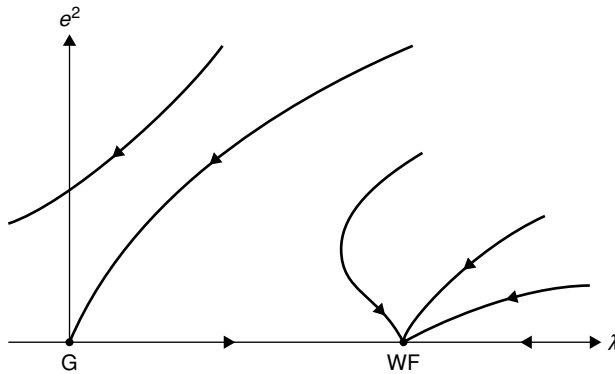


Figure 8.7 Flow on the critical surface for bosons interacting via Coulomb interaction at a commensurate density in dimension $d < 3$. The transition may be continuous or discontinuous depending on the value of the charge.

dimension $d \leq 3$, if the bosons are interacting both via short-range and Coulomb interactions.

Solution At a commensurate density the limited gauge-invariance observed in the previous problem dictates that the action in the presence of a Coulomb interaction between bosons may be written as

$$S = \int d\tau d\vec{x} [(\partial_\tau - ia(\vec{x}, \tau))\Phi(\vec{x}, \tau)|^2 + |\nabla\Phi(\vec{x}, \tau)|^2 - \mu|\Phi(\vec{x}, \tau)|^2 + \lambda|\Phi(\vec{x}, \tau)|^4 + \frac{1}{2e^2}a(\vec{x}, \tau)|\nabla|^{d-1}a(\vec{x}, \tau),$$

in d dimensions. Integrating the fast modes with $\Lambda/b < k < \Lambda$ and $-\infty < \omega < \infty$ leads to changing the coefficients in front of ω (Z_ω), k^2 (Z_k), μ (Z_μ), λ (Z_λ), and a -term (Z_a) in the Fourier transformed action. One then rescales the momentum as $bk \rightarrow k$, the frequency as $b^z\omega \rightarrow \omega$, and the fields as $b^{-d}a \rightarrow a$ and $b^{-(2+d+z)/2}Z_k^{1/2}\Phi \rightarrow \Phi$ to find finally that the action at the critical point can be restored into its original form provided that the coupling constants are changed into $\lambda(b) = b^{4-d-z}Z_k^{-2}Z_\lambda\lambda$ and $e^2(b) = b^{z-1}Z_a^{-1}e^2$. Computing the Z -factors to the leading order then yields

$$\beta_\lambda = \left(\epsilon + \frac{e^2}{2}\right)\lambda - \frac{5}{2}\lambda^2 - \frac{1}{4}e^4,$$

$$\beta_e = (z-1)e^2 - \frac{\delta_{3,d}}{12}e^4,$$

where $\epsilon = 3 - d$. The dynamical exponent is fixed by requiring that $b^{-2z} Z_\omega = b^{-2} Z_k$, so that

$$z = 1 - \frac{1}{3}e^2.$$

The last term in β_e is finite only when $d = 3$, and the quadratic term for the gauge field becomes analytic in momentum.

Small charge is therefore irrelevant at the Wilson–Fisher critical point at $e^2 = 0$ and $\lambda = 2\epsilon/5$, when $\epsilon > 0$. The transition for weak charge therefore remains continuous, in the XY universality class, and with $z = 1$. For large charge, on the other hand, the transition becomes fluctuation-induced first order. This follows from observing that the flow originating on the line $\lambda = 0$ is towards negative λ , so there must exist a separatrix that divides the flow diagram into the regions of first and second order transitions. This situation is depicted in Fig. 8.7.

Finally, precisely in $d = 3$ one can find the flow trajectories explicitly:

$$C + \ln e^2 = \frac{60}{239} \arctan \left(\frac{60(\lambda/e^2) - 11}{\sqrt{239}} \right),$$

where C is a constant. All solutions run away to the region with negative λ , and the transition is consequently first order for all values of the parameters.

